

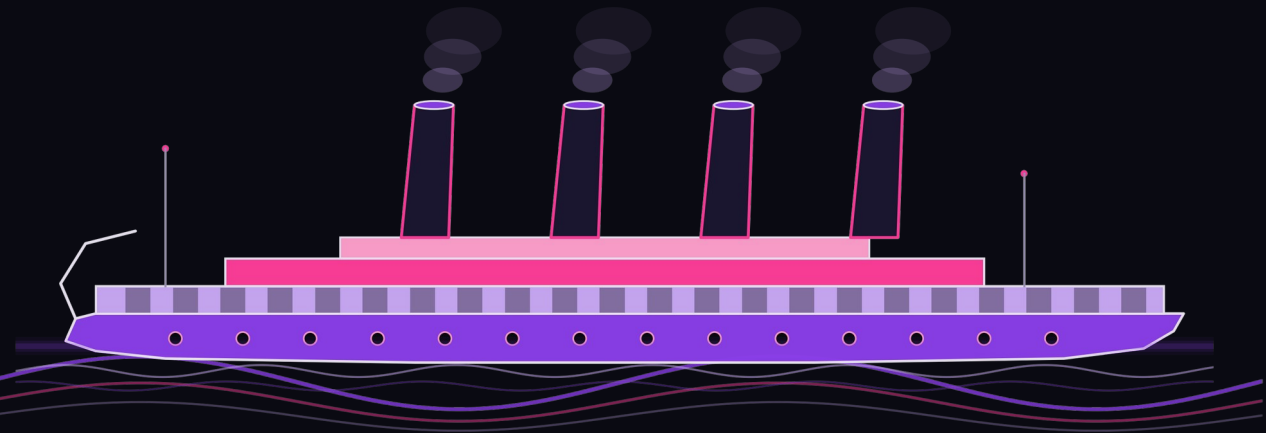
[DATA ANALYST // UNIT SUMMARY PROJECT]

TITANIC

Decoding who survived the wreck — and why

RITA LIMA

Correlation · Research Questions · Machine Learning





Mission Briefing

01

Data Preparation

Cleaning, missing values, and outlier handling

02

Exploratory Analysis

Distributions by gender, age, and survival

03

Correlation

What signals actually relate to survival?

04

Research Questions

Three questions explored in depth

05

Machine Learning

Three models trained to predict survival

891

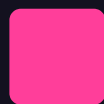
passengers in dataset

38.4%

overall survival rate



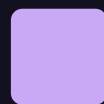
Cleaning the Dataframe



Cabin

77.1%

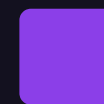
missing — column dropped; not reliable enough to reconstruct



Age

19.9%

missing — filled with median age, split by gender



Embarked

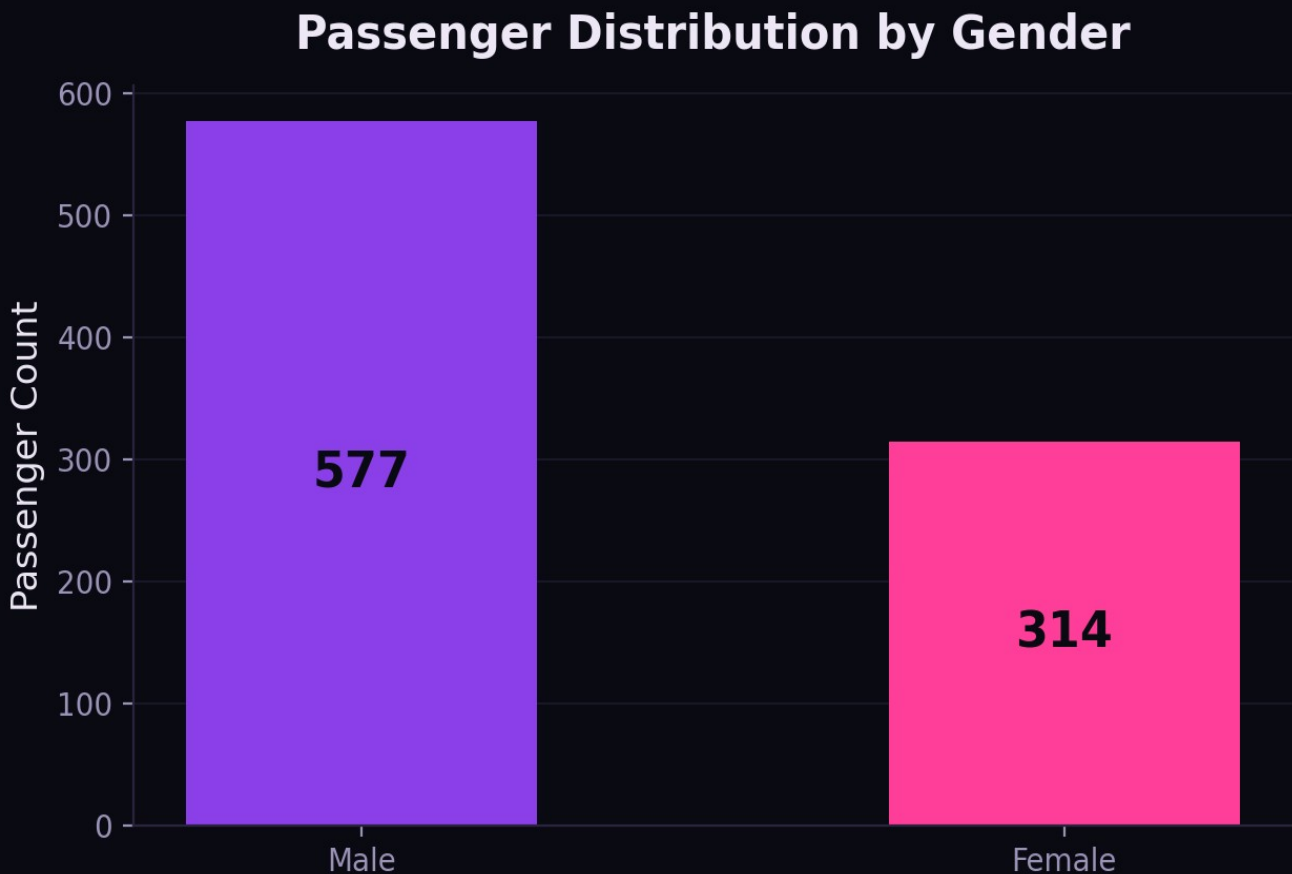
0.2%

missing — filled with the most frequent port (mode)

METHOD NOTE: medians were computed separately for men and women, since age distributions differ by gender — this avoids biasing the imputation.



Gender Distribution



577

male passengers (64.8%)

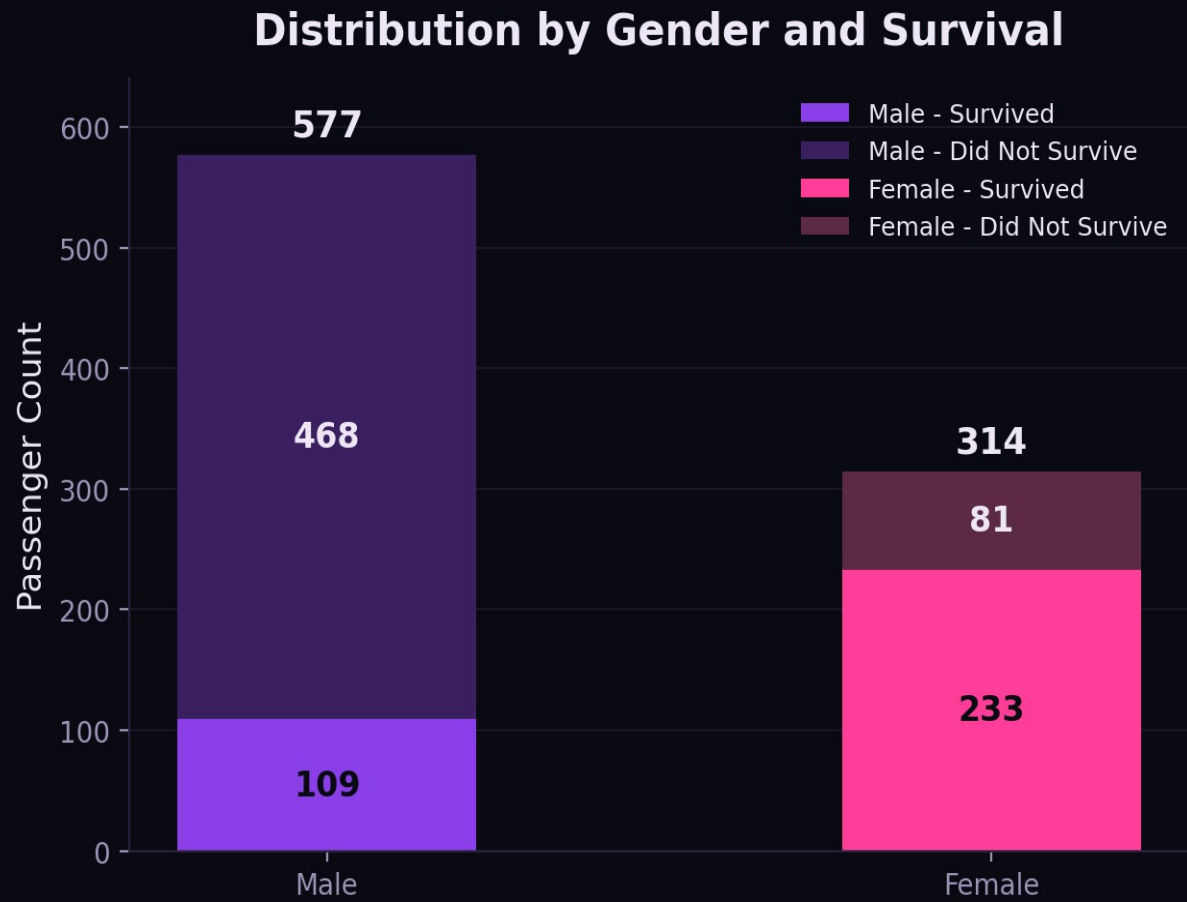
314

female passengers (35.2%)

The dataset skews almost 2:1 male — a gap worth keeping in mind for every gender comparison ahead.



Gender and Survival



74.2%

of women survived

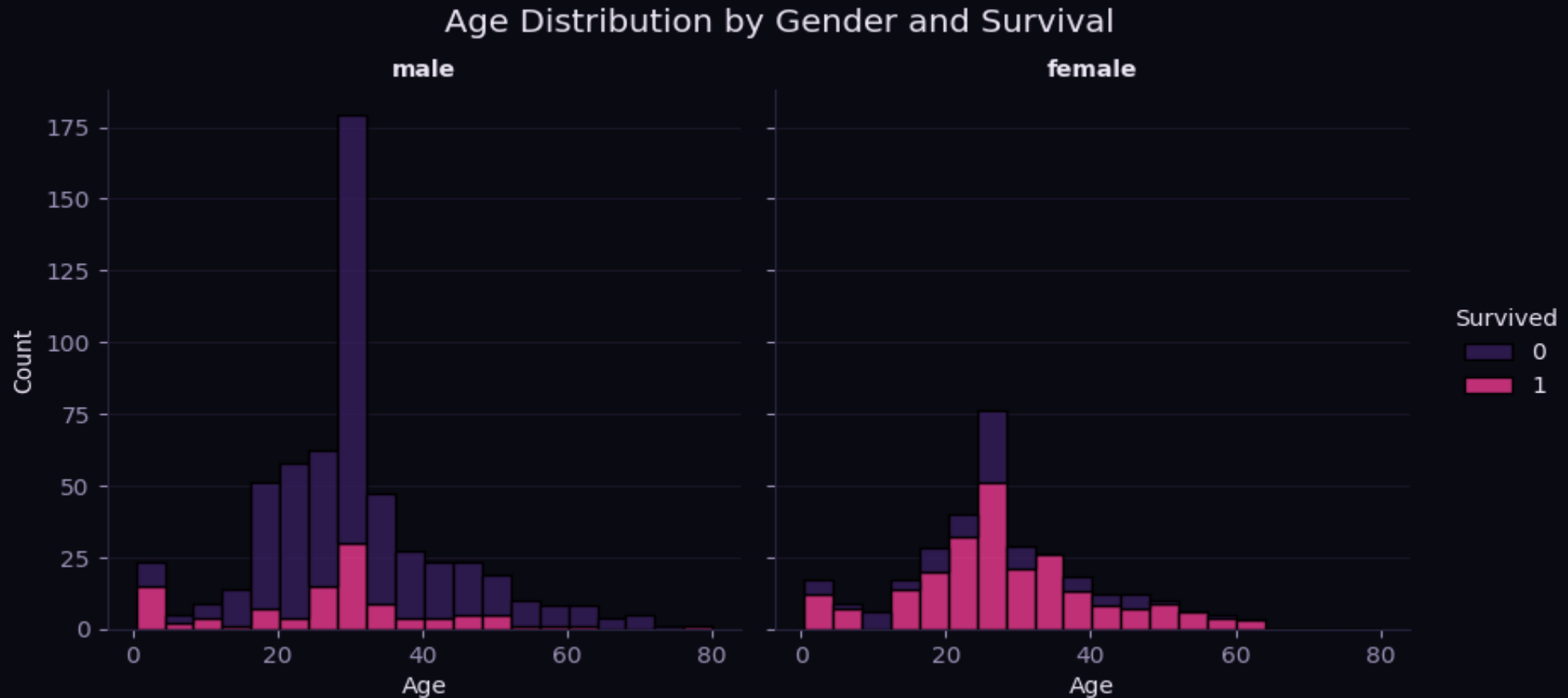
18.9%

of men survived

A near 4x gap points to a strong “women and children first” protocol effect.



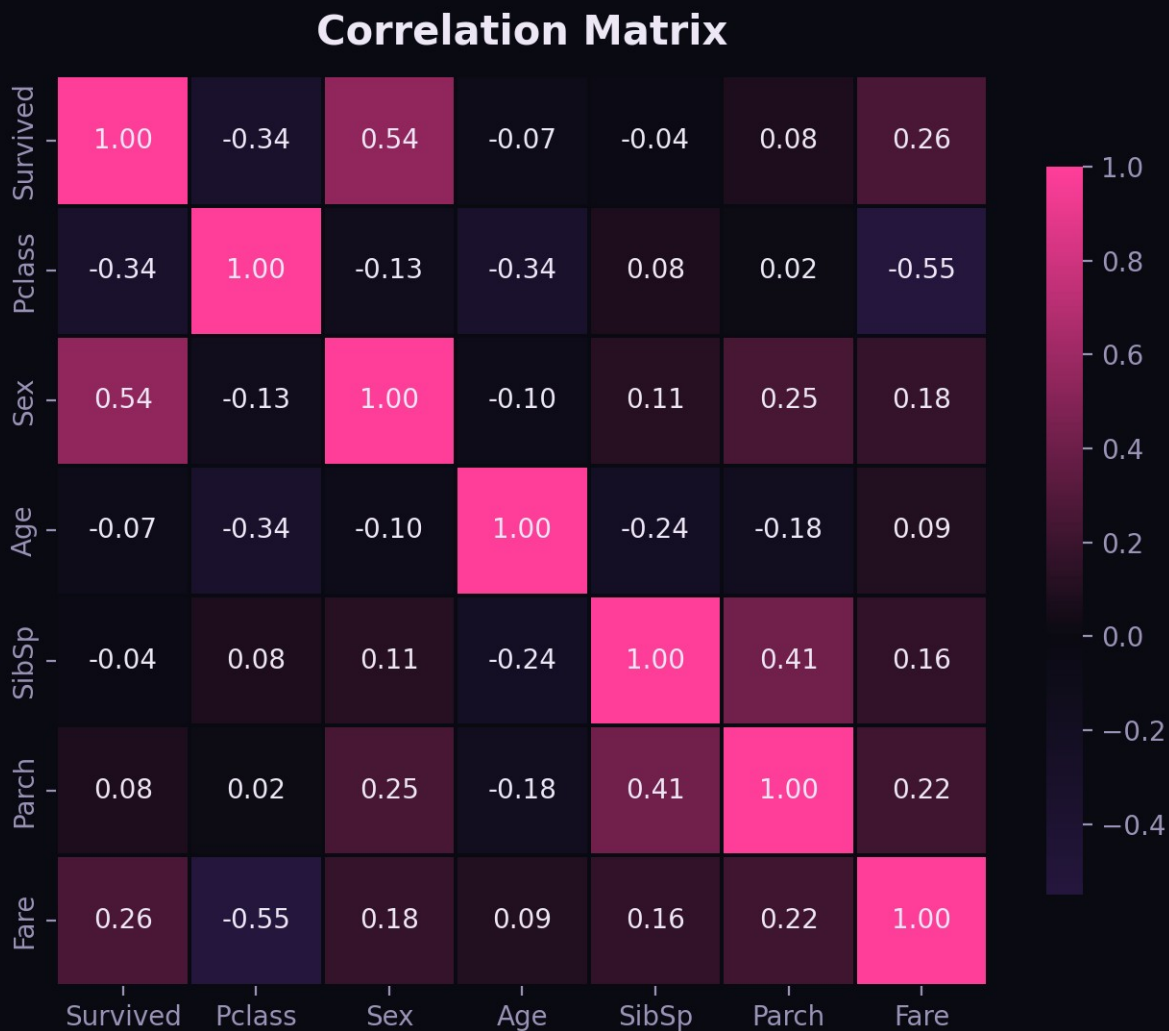
Age Distribution



In both genders, survival concentrates at younger ages — the effect reads more clearly among men.



Correlation Matrix



SIGNAL READOUT

- +0.54** Sex ↔ Survived
moderate positive correlation — being female associates with higher survival
- 0.34** Pclass ↔ Survived
lower class (higher number) associates with lower survival
- 0.55** Pclass ↔ Fare
the strongest correlation in the matrix — better classes pay higher fares
- 0.07** Age ↔ Survived
near zero — age alone barely explains survival



Sex and Class, Combined



96.8%

of 1st class women survived

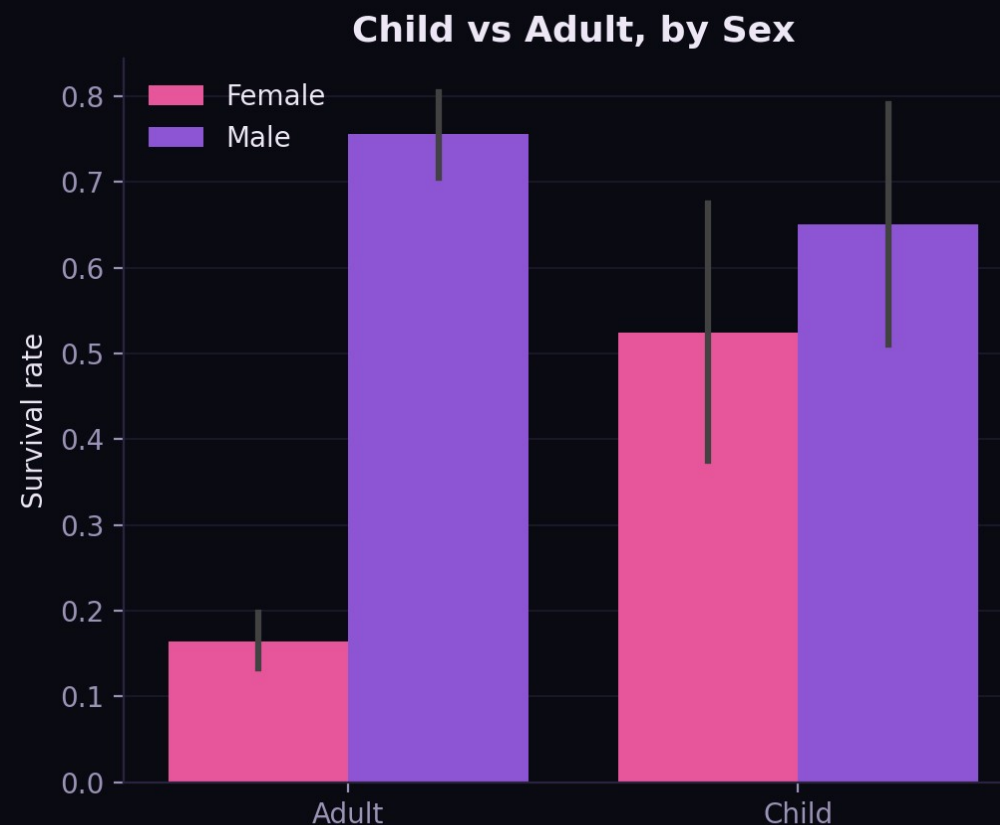
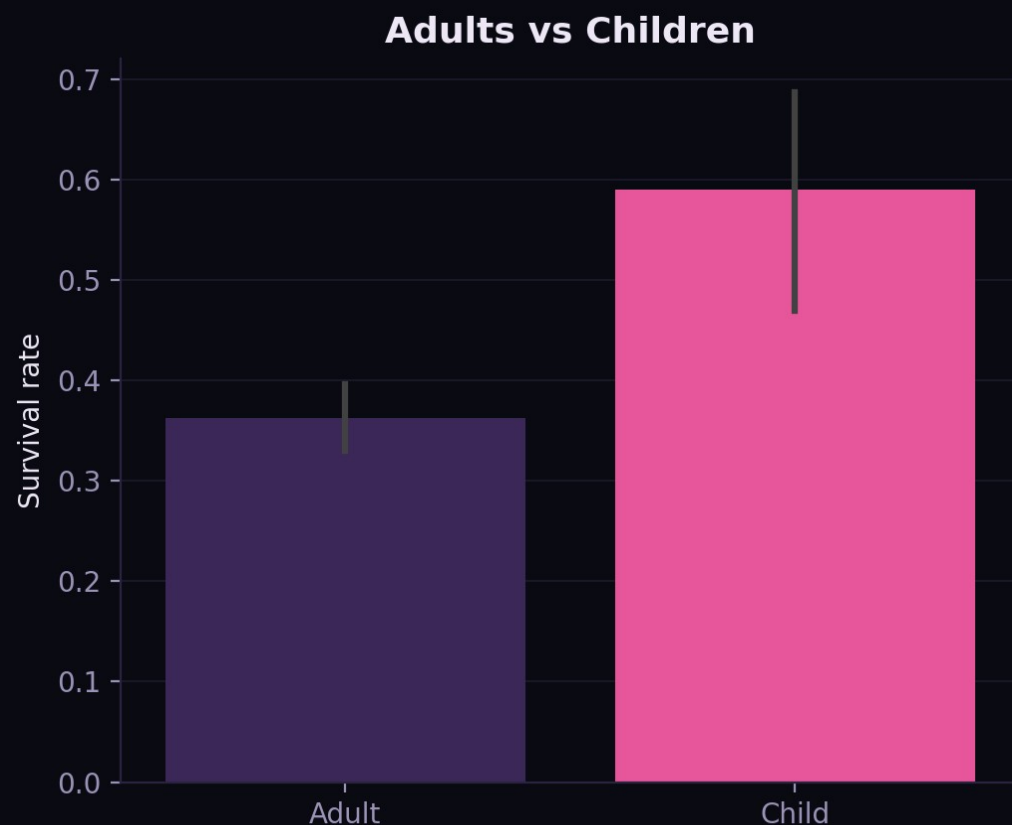
13.5%

of 3rd class men survived

Gender dominates the overall effect, but class clearly modulates it within each gender — especially among women, whose survival rate drops from 96.8% (1st class) to 50.0% (3rd class).



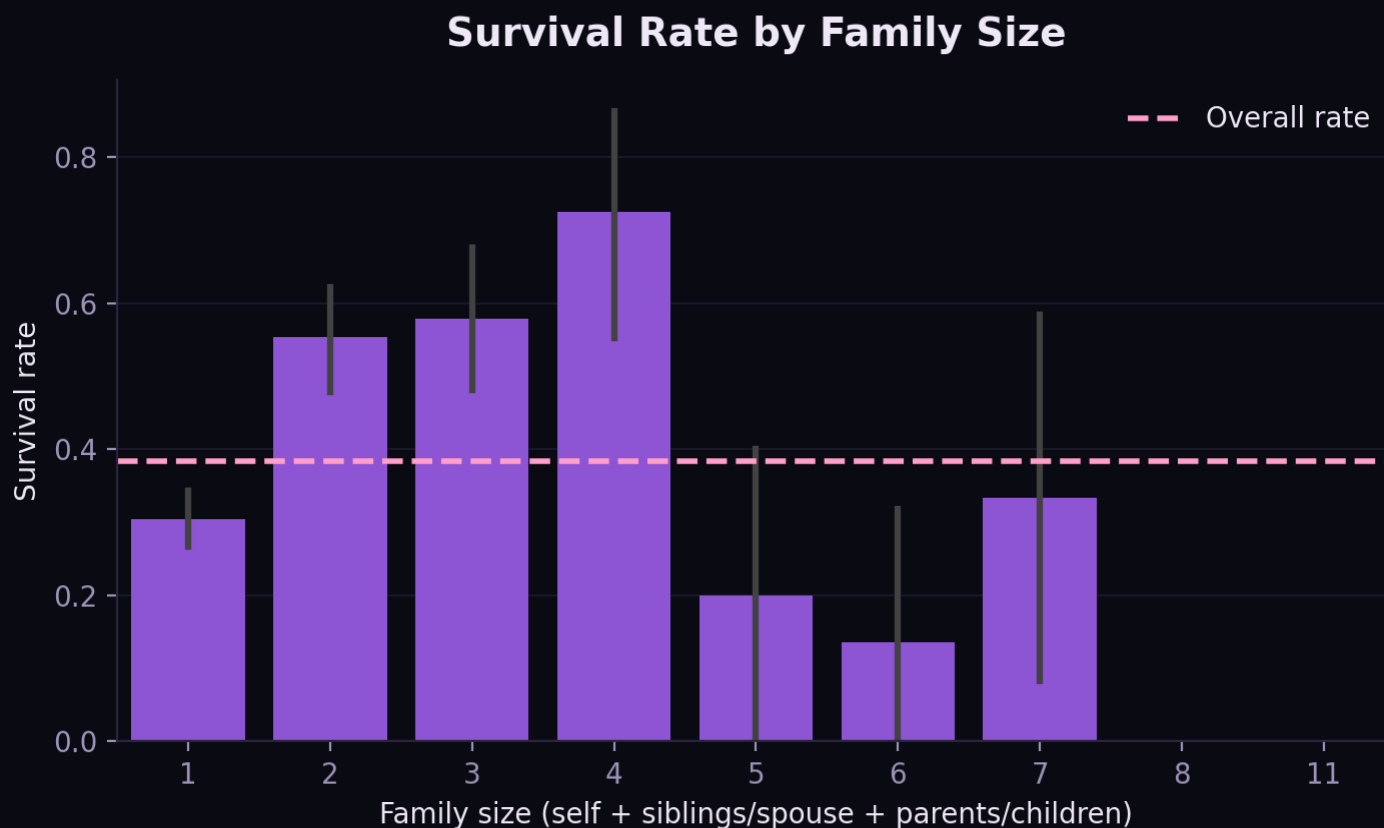
The Age Effect on Children



Children (< 16) survive more than adults (59.0% vs 36.3%) — but the effect isn't equal by gender: for boys, being a child nearly triples the survival rate (52.5% vs 16.4%), while for girls the effect is small and even slightly reversed.



Traveling Alone vs. in a Family



NON-LINEAR PATTERN

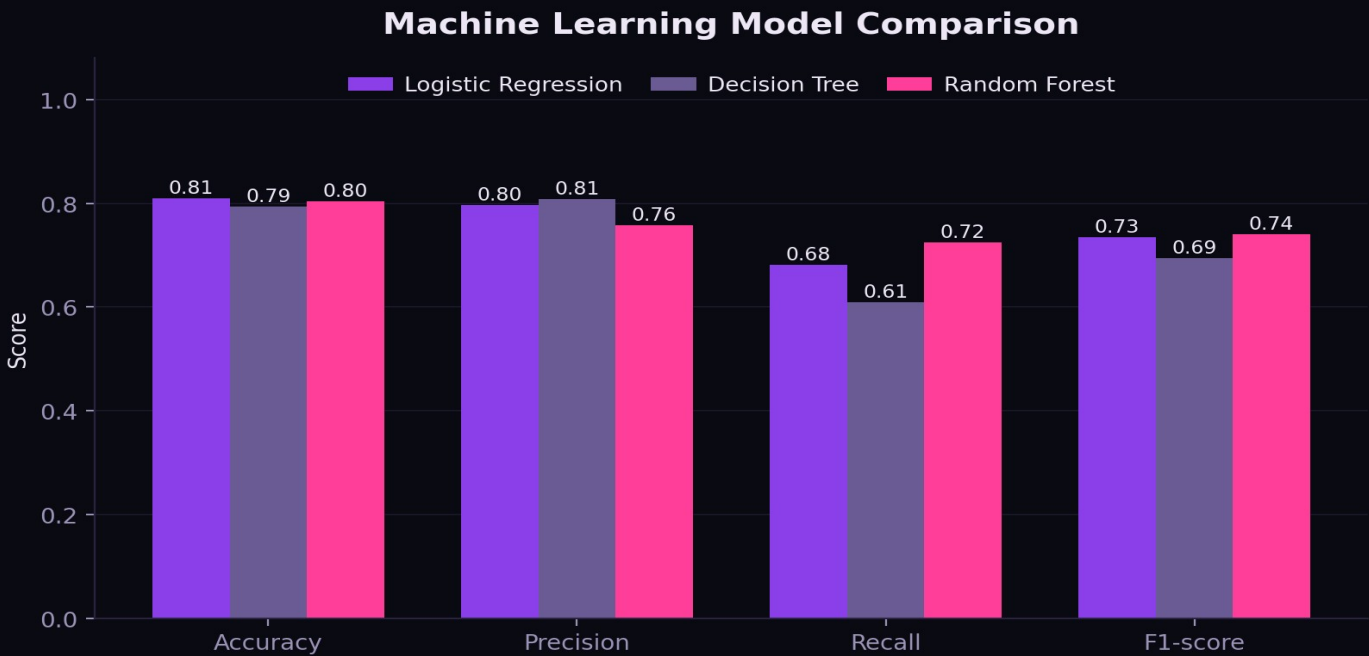
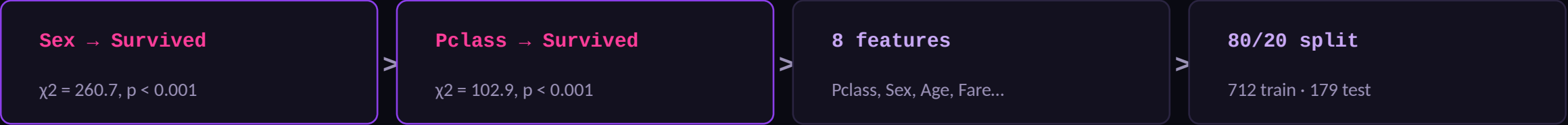
Solo travelers: 30.4% survival — the lowest value
Family of 2 to 4: 55.3% to 72.4% — the sweet spot
Family of 5+: survival drops again (13.6% to 33.3%)

Neither traveling solo nor in large groups favors survival — small families could coordinate without being a burden during evacuation.



Predicting Survival

Signal validated, pipeline built, three models put to the test.



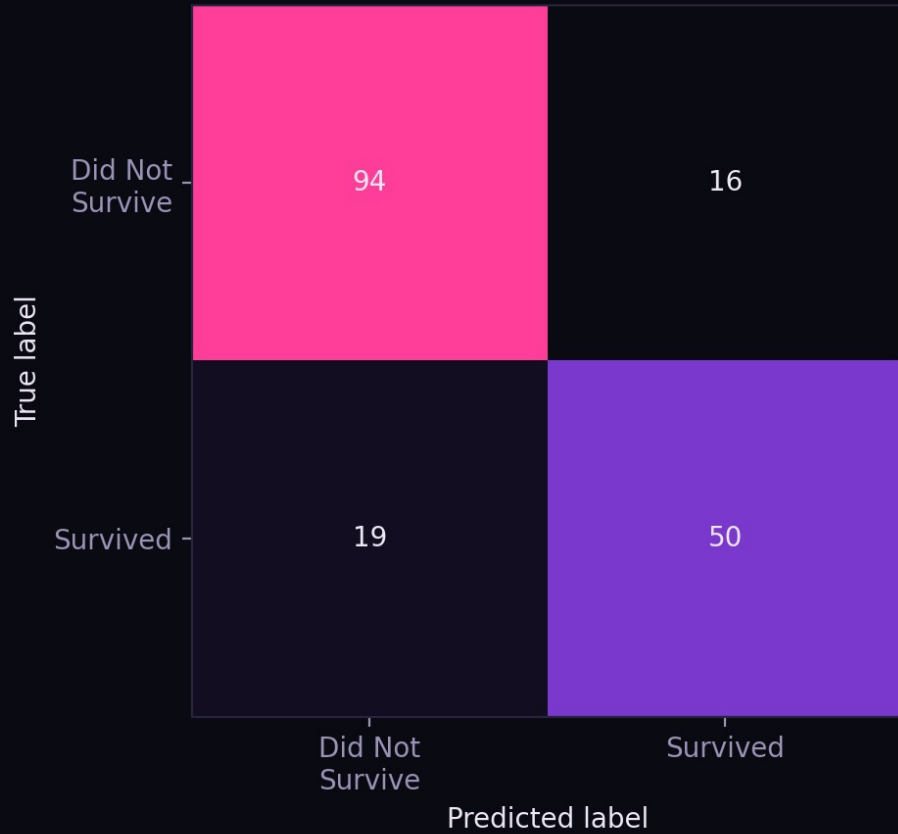
METRICS GUIDE

- **Accuracy** — share of all predictions that were correct
- **Precision** — of predicted survivors, % who actually survived
- **Recall** — of actual survivors, % the model correctly found
- **F1-score** — balance between precision and recall

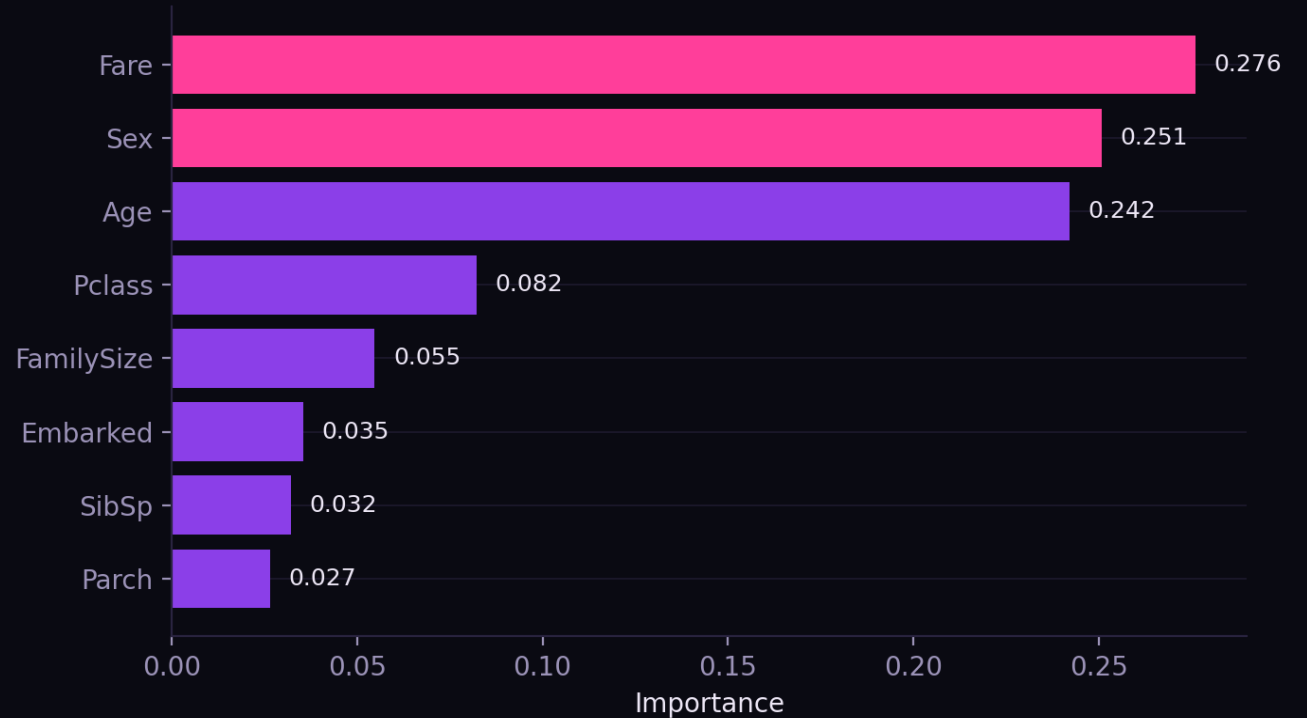
Random Forest wins: best balance of precision and recall, with the highest F1-score (0.741).

Random Forest, In Detail

Random Forest — Confusion Matrix



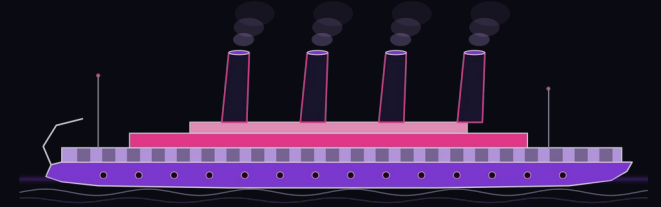
Random Forest — Feature Importance



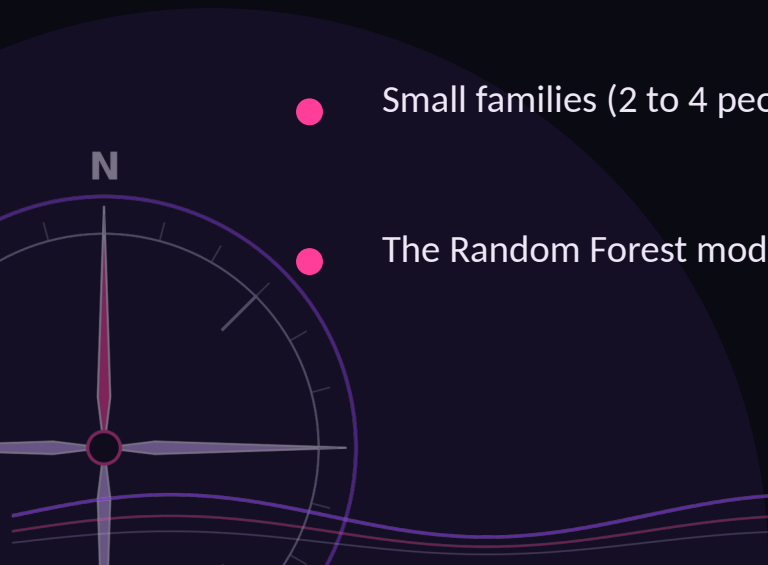
Fare, Sex, and Age dominate the model's decisions — reinforcing what the correlation matrix and research questions already signaled.

[CONCLUSION]

Who Survived the Titanic?



- Gender was the single strongest factor — women had nearly 4x the survival rate of men.
- Ticket class amplified or dampened that effect, especially among women.
- Being a child helped boys far more than girls, whose survival was already high.
- Small families (2 to 4 people) outlived both solo travelers and large groups.
- The Random Forest model confirms these patterns: Fare, Sex, and Age are the top features.



// end of transmission